Experiment 1

DFT and Its Properties

Lab Work

a. Load the image file barbara_gray_512.bmp (You would use MATLAB command 'imread' to load this file to a variable 'test_image'. Use MATLAB help to learn the usage of 'imread', then check for 'imwrite'). This file contains an image with several frequency components.
b. To observe the frequency components, use fft2 function, then calculate the magnitude of the Fourier transformed image. Afterwards, to improve the visibility of the magnitude pass it through log10 function. Diplay the result.
c. Downsample 'test_image' by 2. Therefore size of th image will be 256 by 256 at the end. Then, display it and save in bmp format. As in part (b), observe its frequency components.
d. Upsample the image you obtain in part (c). Again, display it and save it in bmp format. Furthermore, observe its frequency compoents.
e. Find a method to recover the original image (Hint: you can use pre-intalled interpolation methods in MATLAB). Check its frequency components.
f. Discuss the result you obtain in part (e)